

Dynamic Split-Testing Simulator

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Subject : React JS

Problem Statement: Dynamic Split-Testing (A/B) Landing Page Simulator

Github Repository Link:

<https://github.com/sejaltech/Dynamic-Split-Testing-Simulator.git>

Live Demo Link: <https://glowing-madeleine-8013db.netlify.app/>

Project Title

Dynamic Split-Testing Simulator

A React-based A/B Testing Simulator that enables users to configure experiments, simulate visitor traffic, analyze variant performance, visualize conversion trends, and determine winning variants using Frequentist and Bayesian statistical techniques.

Problem Statement

Modern digital products rely heavily on A/B testing to optimize user engagement, conversion rates, and business outcomes. However, commercial A/B testing platforms often require backend infrastructure, databases, analytics services, and tracking systems, making them difficult to understand and expensive to implement.

The objective of this project is to develop a frontend-only Dynamic Split-Testing Simulator that replicates the behavior of professional experimentation platforms. The system allows users to create multiple variants, allocate traffic, simulate thousands of visitors, calculate statistical confidence levels, and visualize experiment outcomes in real time without requiring any server-side infrastructure.

The simulator helps students, marketers, product managers, and developers understand A/B testing methodologies through an interactive and data-driven environment.

TECH STACK

Frontend Technologies

- ReactJS
- Vite
- JavaScript (ES6+)
- HTML5
- CSS3

State Management

- Zustand

Used for centralized experiment state management and efficient UI updates.

Data Visualization

- Chart Library (used for analytics visualizations)
- Real-time performance charts
- Conversion trend graphs

Storage

- Browser Local Storage
- JSON Export / Import

Deployment

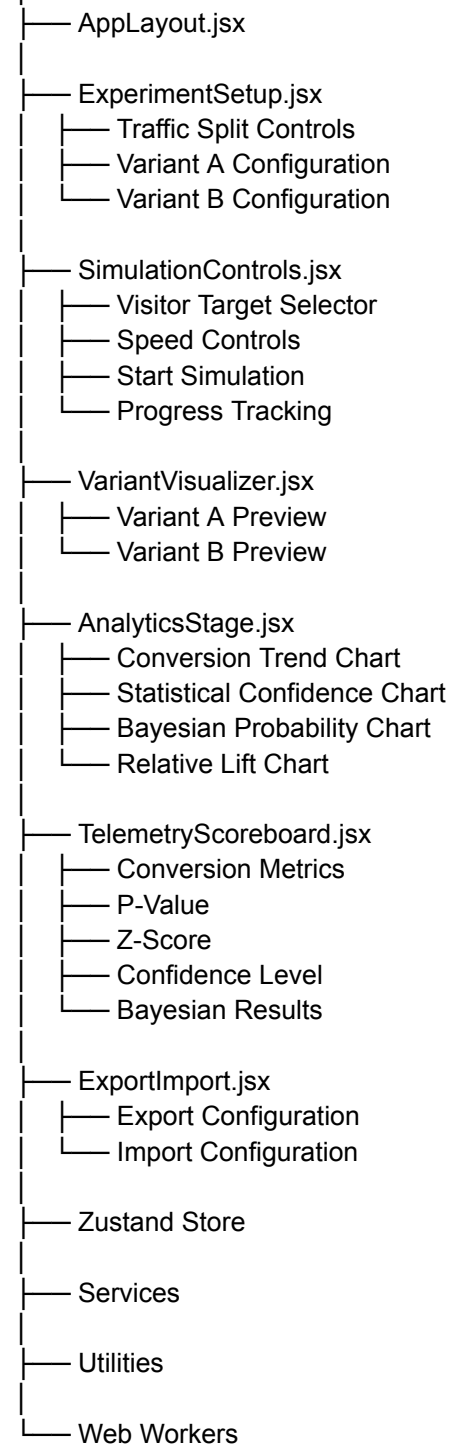
- Netlify

Version Control

- Git
- GitHub

COMPONENT ARCHITECTURE

App.jsx



FEATURES IMPLEMENTED

1. Experiment Configuration

Users can:

- Configure Variant A and Variant B
- Set variant names
- Modify landing page headlines
- Change CTA button text
- Select button colors
- Adjust font sizes
- Configure true conversion rates

2. Traffic Split Simulation

The simulator provides a traffic allocation slider that dynamically distributes visitors between:

- Variant A
- Variant B

This mimics real-world traffic splitting used in production A/B tests.

3. Monte Carlo Simulation Engine

The project uses probabilistic simulation techniques to generate virtual visitors.

For each visitor:

1. A variant is selected.
2. Conversion probability is evaluated.
3. Conversion outcome is generated.
4. Metrics are updated in real time.

4. Simulation Controls

The platform includes:

- Visitor Target Selection
- Speed Adjustment
- Progress Tracking
- Start Simulation

allowing users to run large-scale experiments interactively.

5. Variant Visualization

A live preview panel displays both variants simultaneously.

Users can instantly observe:

- Headline changes
- CTA text changes
- Button color modifications
- Layout differences

This provides immediate visual feedback during experimentation.

6. Real-Time Analytics Dashboard

The analytics dashboard continuously updates:

- Conversion Rates
- Conversion Counts
- Relative Lift
- P-Value
- Z-Score
- Frequentist Confidence

allowing users to monitor experiment performance.

7. Bayesian Analysis

The simulator includes Bayesian probability calculations that estimate:

- Probability that Variant A beats Variant B
- Probability that Variant B beats Variant A
- This provides a more intuitive interpretation of experiment outcomes.

8. Statistical Significance Evaluation

The application calculates:

- Statistical Confidence
- Relative Lift
- Significance Levels

helping determine whether observed differences are meaningful.

9. Data Visualization

Interactive charts display:

- Conversion Rate Trend
- Statistical Confidence Progression
- Bayesian Win Probability
- Relative Lift Progression

making experiment data easier to understand.

10. Export & Import Support

Users can:

- Export experiment configurations
- Save simulation setups
- Import previously saved experiments

using JSON-based data exchange.

SCREENSHOTS

Figure 1: Main Dashboard

Description:

The main dashboard provides experiment configuration, simulation controls, and traffic allocation management.

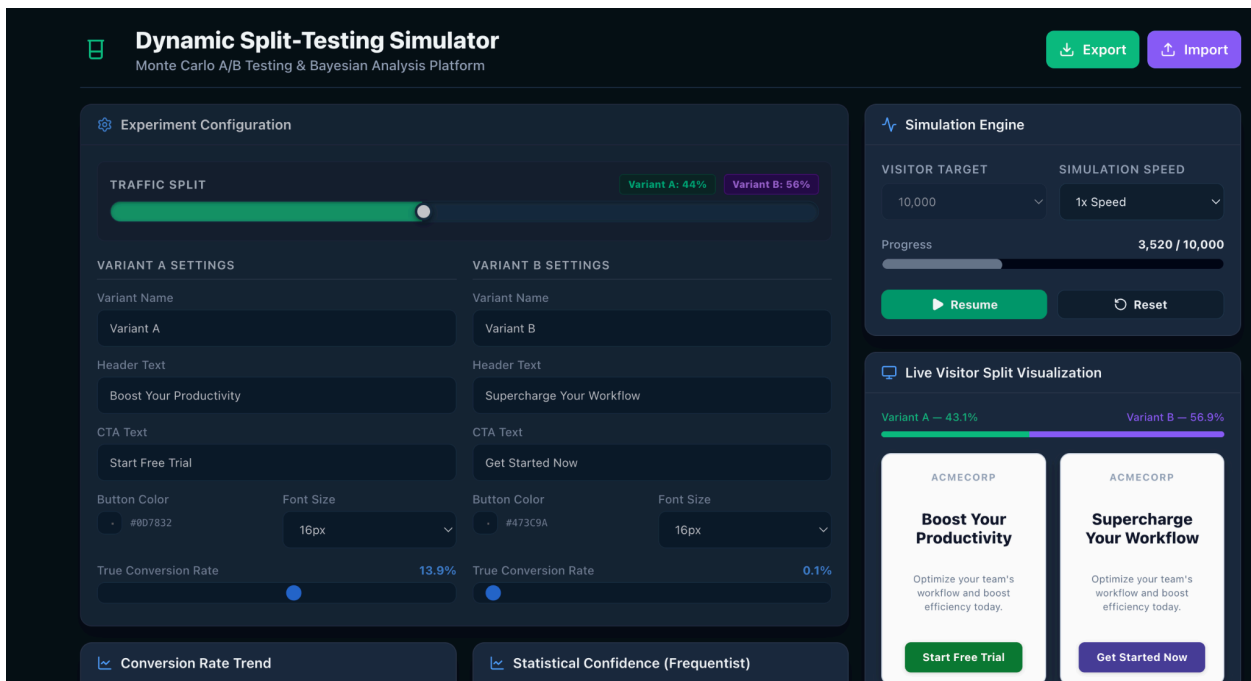


Figure 2: Analytics & Statistical Analysis

Description:

Displays conversion trends, confidence calculations, Bayesian probabilities, and relative lift progression charts.

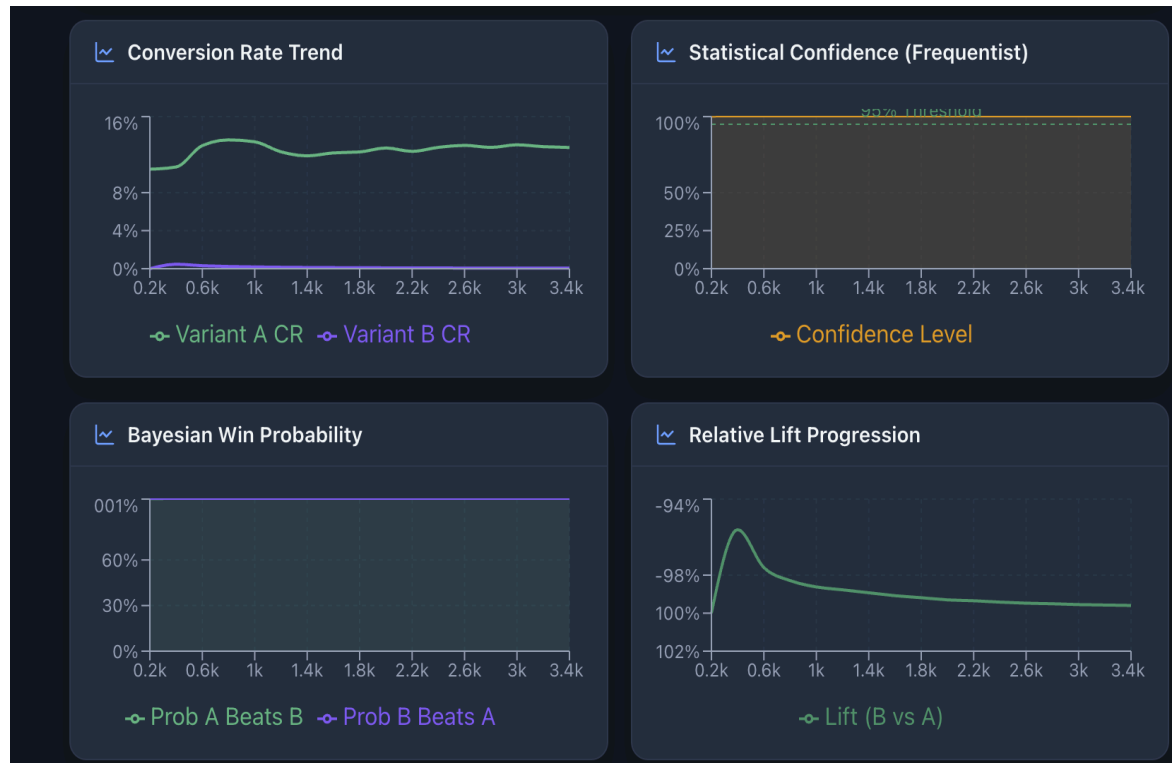
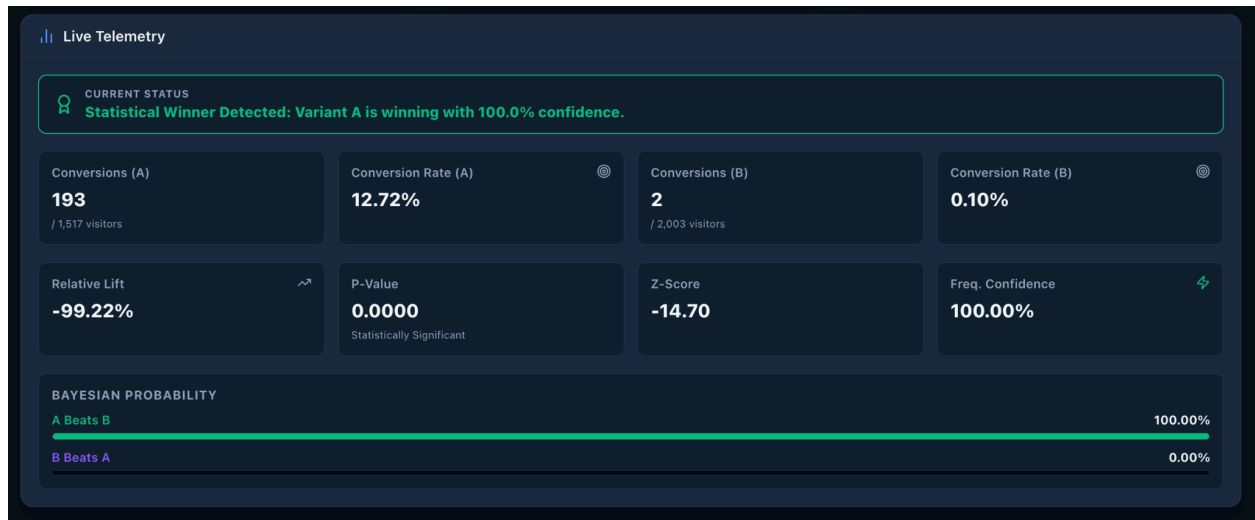


Figure 3: Live Telemetry Dashboard

Description:

Shows real-time conversion metrics, p-values, z-scores, confidence levels, and Bayesian winner probabilities.



CONCLUSION

The Dynamic Split-Testing Simulator successfully demonstrates the core principles of A/B testing through an entirely frontend-based implementation. Using ReactJS, Zustand state management, Monte Carlo simulation, Bayesian probability analysis, and statistical significance calculations, the application provides a realistic experimentation environment without requiring backend infrastructure.

The project enables users to configure landing page variants, simulate visitor behavior, monitor conversion performance, and analyze statistical outcomes through interactive dashboards and visualizations. Additionally, the export/import functionality and centralized state architecture make the system scalable and maintainable.

Overall, the project effectively showcases modern frontend development practices while providing practical insights into experimentation platforms used in real-world product development and digital marketing.

THANK YOU